CentOS Stream 10 & RHEL 10: Local Repository Configuration Guide

A hands-on lab reference guide for building isolated, offline package delivery systems.

Step 1: Create a Permanent Directory

```
mkdir -p /media/cdrom
```

What we are doing: Creates a new directory named **cdrom** inside the **/media** folder. The **-p** flag safely creates any missing parent directories.

Why we are doing it: The default GUI automount paths are temporary and session-dependent. This creates a fixed, structural anchor point on the file system.

Step 2: Mount the ISO File Permanently

```
mount -o ro /dev/sr0 /media/cdrom
```

What we are doing: Links the virtual optical media hardware device (**/dev/sr0**) to the structural directory target (**/media/cdrom**) under Read-Only (**-o ro**) access permissions.

Why we are doing it: Linux treats hardware components as block devices. Mounting links the system's storage sectors to an accessible directory structure.

Step 3: Create the Local Repository Configuration File

```
cat << 'EOF' > /etc/yum.repos.d/local.repo
[local-baseos]
name=CentOS Stream 10 - Local - BaseOS
baseurl=file:///media/cdrom/BaseOS
enabled=1
gpgcheck=1
gpgkey=file:///media/cdrom/RPM-GPG-KEY-centosofficial

[local-appstream]
name=CentOS Stream 10 - Local - AppStream
baseurl=file:///media/cdrom/AppStream
enabled=1
gpgcheck=1
gpgkey=file:///media/cdrom/RPM-GPG-KEY-centosofficial
EOF
```

What we are doing: Constructs a new repository definition configuration mapping profile (**local.repo**) inside the system directory **/etc/yum.repos.d/**.

Why we are doing it: DNF continuously monitors this directory path for target package paths. Setting **baseurl=file:///** routes the manager locally. **gpgcheck=1** guarantees file authenticity against the embedded signature keys.

Step 4: Disable Online Configuration Profiles (True Offline Simulation)

```
sed -i 's/enabled=1/enabled=0/g' /etc/yum.repos.d/centos.repo
```

What we are doing: Applies a stream text editor (**sed**) command inline (**-i**) to globally replace operational statuses from enabled to disabled inside the default system configurations.

Why we are doing it: If internet interfaces are dropped or restricted, DNF fails continuously while trying to reach external web mirrors. Disabling them forces reliance solely on your offline media assets.

Step 5: Flush Package Infrastructure Cache

```
dnf clean all
```

What we are doing: Deletes all downloaded operational state metadata databases, headers, and historical package map caches stored on the server disk files.

Why we are doing it: Forces DNF to abandon stale metadata memory schemas, preparing it to process your newly registered offline path maps accurately.

Step 6: Build Fresh Metadata Caches

```
dnf makecache
```

What we are doing: Scans your configured offline target directory endpoints, pulls package headers, and creates a local index cache database.

Why we are doing it: Acts as an end-to-end configuration check, confirming DNF can map packages from the local directory safely.

Step 7: Verify Repository Active Status Maps

```
dnf repolist
```

What we are doing: Requests an environment summary display of all active package storage architectures recognized by the manager engine.

Why we are doing it: Provides immediate functional verification. Successfully configured labs display **local-baseos** and **local-appstream** in active operational status.

Step 8: Automate Mount States Across System Restarts

```
echo "/dev/sr0 /media/cdrom iso9660 defaults,ro 0 0" >> /etc/fstab
```

What we are doing: Appends persistent mount mappings to the system file-system table file (**/etc/fstab**).

Why we are doing it: Ensures that the server reattaches the ISO media workspace to its correct destination automatically whenever the server reboots.

This is for informational purposes only. AI responses may include mistakes.